

Lab: Signal Taxonomy and Failure Analysis

Objective

Practice reading prototype signals by classifying test data into signal types, interpreting failure modes as informative events, and designing instrumentation to extend your perceptual reach. By the end of this lab, you will have a signal collection plan for your invention project.

Prerequisites

- Familiarity with observation vs. inference, signal types, and failure interpretation from this module's lecture
- Your invention project prototype or prototype specification from Module 01
- A test scenario (real or simulated) where you can generate or imagine prototype data

Materials

- Notebook or digital document
- Your prototype (or a detailed description if the physical prototype is not available)
- Colored pens or highlighters (for categorizing signals)

Part 1: Observation Before Inference (10 minutes)

Goal: Practice separating raw observations from interpretations.

Conduct a brief test of your prototype (or simulate one by closely examining your prototype specification and imagining a test). Record observations in two columns: what you actually observed (raw data) and what you interpreted (inference).

Raw Observation (What you saw/measured/heard)	Your Inference (What you think it means)	Could another interpretation be valid?

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Reflect: How easy or difficult was it to separate observation from inference? Did you catch yourself interpreting before recording?

Write your response here

Part 2: Signal Classification (15 minutes)

Goal: Classify the signals your prototype could generate using the four-category taxonomy.

For your invention project, brainstorm all the signals a test could produce. Then classify each:

Quantitative Measurements — things you could measure with a number:

Measurement	Unit	What it tells you	How precise does it need to be?

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Qualitative Feedback — things users or experts might say:

Possible feedback	What it would suggest about your model	Parameter update or structure update?

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Behavioral Observations — things you would see users do:

Possible behavior	What it would reveal about the user's generative model	How would you record it?

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Potential Failure Events — ways the prototype could fail:

Failure mode	What it would reveal	Severity (minor/major/critical)

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Part 3: Failure Mode Deep Dive (15 minutes)

Goal: Treat failure as the most informative signal by analyzing a specific failure mode in detail.

Select one potential failure mode from Part 2 (or recall an actual failure if you have tested a prototype). Analyze it thoroughly:

The failure:

Write your response here

When it occurs: Under what conditions does this failure happen? What triggers it?

Write your response here

What your generative model predicted: What did you expect to happen instead?

Write your response here

The prediction error: What is the specific discrepancy between prediction and observation?

Write your response here

Root cause analysis — 5 Whys:

Level	Why?	Answer
1	Why did this failure occur?	

Write your response here

|| 2 | Why? (deeper) |

Write your response here

|| 3 | Why? (deeper) |

Write your response here

|| 4 | Why? (deeper) |

Write your response here

|| 5 | Why? (root cause) |

Write your response here

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What this failure teaches you: - About the prototype's physics/mechanism:

Write your response here

- About the user's expectations:

Write your response here

- About your generative model's blind spots:

Write your response here

How to update your model based on this failure: - Parameter update (change a value):

Write your response here

- Structure update (add a variable or relationship):

Write your response here

- Do nothing (failure is outside design scope):

Write your response here

Part 4: Instrumentation Design (10 minutes)

Goal: Identify hidden states you cannot currently perceive and design instrumentation to access them.

What aspects of your prototype's behavior are currently invisible to you? What hidden states would you like to observe but cannot with your current perceptual capabilities?

Hidden State	Why it matters	Instrument/method to make it observable	Estimated cost/difficulty

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If your prototype is digital, consider: What user behaviors are you not currently logging? What system states are invisible in your current monitoring setup?

Write your response here

If your prototype is physical, consider: What internal stresses, temperatures, vibrations, or movements are currently unobservable? What sensors could you add?

Write your response here

Part 5: Signal Collection Plan (10 minutes)

Goal: Create a structured plan for collecting and recording signals during your next test.

Signal Collection Plan for:

Write your response here

Signal Category	Specific Signals to Collect	Collection Method	Recording Format	Who Collects
Quantitative				

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|| Qualitative |

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| | **Behavioral** |

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|| Failure Events |

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Data separation protocol: How will you ensure raw observations are recorded before interpretations?

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Part 6: Cross-Signal Integration (5 minutes)

Goal: Practice combining signals from different categories to form a holistic picture of your prototype's performance.

Look across all the signals you identified in Parts 2-5. Do any signals from different categories tell a consistent story? Do any contradict each other?

Story / Pattern	Supporting Signals (list category and specific signal)	Contradicting Signals	Your Interpretation

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When signals converge (multiple signal types pointing to the same conclusion), your confidence in the interpretation should increase. When signals diverge (quantitative data says one thing, qualitative feedback says another), this is a flag for deeper investigation — the divergence may reveal that your generative model is structured incorrectly.

What is the single most important signal your testing has produced so far? Which category does it belong to?

Write your response here

Discussion and Debrief

1. **Observation discipline:** Was separating observation from inference harder than expected? What does this tell you about how tightly coupled perception and interpretation are?
2. **Signal priorities:** Of the four signal types, which is most important for your invention at its current stage? Why?
3. **Failure reframing:** Did the failure analysis exercise change how you feel about potential failures? Can you honestly say you would welcome a major failure during testing?
4. **Instrumentation gaps:** What is the most important hidden state you identified that you currently cannot observe? How would access to this state change your testing strategy?
5. **Qualitative vs. quantitative:** If you could collect only one type of data in your next test, which would you choose and why? What would you lose by not collecting the other?

Write your response here